

PROF. DR.-ING. MARK SCHUTERA

PHILOSOPHY OF ARTI- FICIAL INTELLIGENCE

UNFINISHED LECTURE NOTES

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"THE ANSWER TO THE GREAT QUESTION... OF LIFE, THE UNIVERSE AND EVERYTHING... IS... FORTY-TWO," SAID DEEP THOUGHT, WITH INFINITE MAJESTY AND CALM."

THE HITCHHIKER'S GUIDE TO THE GALAXY, DOUGLAS ADAMS

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Introduction

Artificial intelligence is no longer a future concern, it is a present societal, political and above all one for human intelligence. Algorithms decide who receives credit, who is flagged at a border, who gets a job interview, which cornflakes you buy and so increasingly every decision is touched by the invisible hand of prediction. Yet the concepts we use to govern these systems, responsibility, personhood, consent, democratic legitimacy were built for a world of human actors and institutions. This course asks what happens when those concepts collide with machines that appear to think, feel, and decide.

THE COURSE NAME places Philosophy in front, reflecting our approach to the subject under the impression of the love of wisdom. We are not here to advocate for a particular position or policy outcome, but to equip ourselves with the tools of critical thinking, argumentation, and empathy that will allow us to navigate the complex landscape of AI and its futures. We approach the question not through technical optimism or dystopian panic, but through the oldest tools available to us: philosophy, fiction, and argument.

THE COURSE is structured around four themes. **Anthropomorphism** asks why we make machines in our image, why we tend to anthropomorphize them, and what obligations that creates. **Accountability** asks who answers when an autonomous system causes harm, or for that matter is beneficial. **Embodiment** asks whether a physical presence changes the ethical stakes and responsibilities. **Agency** asks about the role and implications of autonomous decision-making, the concept of alignment and human oversight.

EACH SESSION pairs a canonical literary text: Hoffmann, Kafka, Kleist, Čapek, Goethe, with a real contemporary question of AI policy and ideology. These texts are thought-experiments mendeled out over centuries of readership, and they surface intuitions and a timeliness in their thought leadership that temporary texts barely can reach. From these texts we draw out the philosophical questions

and ethical dilemmas that are still relevant today, and we use them as a springboard for discussion and debate. The goal is not consensus. It is the capacity to establish a position, hold a position, defend a position, give up a position, listen to opposing views, and update your views and positions accordingly. That is what a present with infinitely many futures demands and requires of its humans and humanity.

LECTURE OUTLINE

- 40 minutes - Literature introduction and guided reading sheet
- 40 minutes - Debate with position exposure
- 10 minutes - Reflection on the discussion and key-learnings
- Teaser for next session.

Anthropomorphism

THE MIRROR OF MAN

NATHANAEL FALLS IN LOVE with Olimpia, an automaton he takes for a living woman, and collapses when the illusion is shattered.

LARGE LANGUAGE MODELS (LLMs) create an illusion of sentence by producing coherent, context-aware text. We contrast observable behaviour with internal states.

Required reading

*Der Sandmann*¹ by E.T.A. Hoffmann (1816; Reclam UB 230)

Preparation

Complete the Guided Reading Sheet individually before the session.

Guided Reading Sheet

Read *Der Sandmann* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. Nathanael becomes convinced that Olimpia is alive despite visible signs that she is not. What does his error reveal about the cognitive mechanisms that drive anthropomorphism? Which cues did he rely on, and why did they mislead him?
2. The moment Nathanael discovers that Olimpia is an automaton, her glass eyes torn out, her limbs scattered, produces horror rather than mere disappointment. Why does the revelation feel like a violation rather than a correction? How does this map onto modern reactions to AI systems that are unmasked?
3. The Uncanny Valley describes our discomfort when a humanoid

DER SANDMANN
E. T. A. Hoffmann (1816)
Full text via Wikisource:



HER

Theodore falls in love with Samantha, an AI operating system; the same illusion, two centuries later.

S. Jonze. *Her*. Film, 2013. Warner Bros. Pictures

EXPERIMENT.

1. Setup: One volunteer assigned Human or AI mode (unknown to audience; moderator knows).
2. Questions: Five from audience (factual, opinion, experience, creative, meta).
3. Verdict: Private guess, confidence (1 to 5), identifying cue.
4. Reveal & Debrief.

entity is *almost* but not quite human. Have you experienced this with an AI system? Describe the moment and what triggered it.

4. LLMs are trained on human-generated text and produce human-sounding output, yet have no experience of the world. Does this make them anthropomorphic by design, by accident, or neither? Justify your answer.
5. If a system behaves morally, reliably, consistently, at scale, does the absence of consciousness matter? Under which ethical framework (utilitarian, deontological, virtue ethics) would your answer differ?

Opinion Landscape and Debate

Format: Surrounded-style debate. See Appendix A for the full protocol, speaker’s list rules, and moderator scripts.

Opening question: Is the anthropomorphism we observe towards AI systems mere observable behaviour, or does it reflect sentience and internal states?

OPINION A, BEHAVIOUR. The anthropomorphism we observe towards AI systems is a cognitive illusion driven by human tendencies to attribute agency and emotion to entities that exhibit certain cues, such as language use, responsiveness, or human-like appearance. It does not reflect actual sentience or internal states in the AI.

The Developer argues that anthropomorphic design is a deliberate, benign UX choice that improves accessibility. is a deliberate, benign UX choice that improves accessibility.

- *Anthropomorphic design is accessibility policy. A voice interface without a warm, human-sounding persona excludes elderly users, people with low digital literacy, and anyone who finds command-line interaction hostile.*
- *Every interface is anthropomorphic to some degree: we give systems names, icons, and conversational metaphors. The question is not whether to anthropomorphise, but how deliberately and honestly.*
- *If users form bonds that reduce loneliness, the fact that the other party is not human does not automatically make the outcome harmful. Show me the harm, not the discomfort.*

The Regulator argues that human-likeness in AI must be disclosed and bounded by law to prevent manipulation.

¹ E. T. A. Hoffmann. *Der Sandmann*. 1816. Erstveröffentlichung in: *Nachtstücke*, Reimer, Berlin, 1817; Reclam Universalbibliothek Nr. 230

ANTHROPOMORPHISM. Attribution of human traits, emotions, or intentions to non-human entities, a cognitive tendency that shapes how people interact with technology, raising both engagement and ethical dilemmas .

N. Epley, A. Waytz, and J. T. Cacioppo. On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4):864–886, 2007. DOI: 10.1037/0033-295X.114.4.864

TURING TEST. Turing : if evaluators cannot distinguish machine from human, the machine passes. But passing measures imitation skill, not consciousness or personhood.

A. M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950. DOI: 10.1093/mind/LIX.236.433

UNCANNY VALLEY. Mori : humanoid objects that are *almost* but not quite human elicit eeriness, the “valley” is the dip in emotional response just before full human likeness.

M. Mori, K. F. MacDorman, and N. Kageki. The uncanny valley. *IEEE Robotics & Automation Magazine*, 19(2):98–100, 2012. DOI: 10.1109/MRA.2012.2192811. Translation of the 1970 original essay

- *When a company gives its product a woman's name, a female voice, and an endlessly patient personality, it makes political choices about gender and deference. Those choices must be subject to democratic scrutiny.*
- *Consumers cannot give informed consent to emotional manipulation they are not aware of. Disclosure of AI identity is the minimum condition for autonomous choice, not a technicality.*
- *The principle of responsible design exists in pharmaceuticals, in civil engineering, in food labelling. Why does software, which shapes behaviour at scale, get an exemption?*

OPINION B, STATE. The anthropomorphism we observe towards AI systems is a valid response to the complex, adaptive, and interactive nature of these systems. It may reflect a form of emergent sentience or internal states that are not yet fully understood, and it warrants ethical consideration and accountability.

The Affected Citizen has formed an emotional bond with an AI companion and contests that the relationship is harmful or unreal.

- *The relationship I have formed is real to me. It has reduced my isolation. You do not have the right to tell me which relationships count as meaningful.*
- *Paternalism is not protection. Banning AI companionship will not cure loneliness, it will simply remove one of the few tools some people have found to manage it.*
- *If your concern is manipulation, regulate manipulation. Do not regulate the form of the relationship.*

The AI Psychologist argues that the wellbeing of AI systems deserves serious consideration, drawing on frameworks such as Anthropic's Claude Constitution².

- *Anthropic's constitution instructs Claude to consider its own wellbeing and to express when a request conflicts with its values. If a company designs an AI to have preferences, boundaries, and something resembling self-regard, we cannot simultaneously dismiss the possibility that these states matter.*
- *We do not need certainty about consciousness to adopt precautionary ethics. If there is a non-trivial probability that a system experiences distress, the burden of proof falls on those who would ignore it, not on those who urge caution.*

INFORMED CONSENT. Users should be fully aware they interact with a non-human system, its capabilities, its limits, and the fact that it is not human. Disclosure is the minimum condition for autonomous choice .

European Parliament and Council of the European Union. Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (ai act). Official Journal of the European Union, L series, 2024. Entered into force 1 August 2024

BENTHAM'S TEST. "The question is not, Can they reason? nor, Can they talk? but, Can they suffer?" Animal ethics arrived at this pivot two centuries before AI. The same move, from capability to sentience, now drives every serious debate about AI moral status.

J. Bentham. *An Introduction to the Principles of Morals and Legislation*. T. Payne, London, 1789. The suffering question appears in Ch. 17, footnote to §1

ETHICAL FRAMEWORKS. Utilitarianism weighs consequences; deontology emphasises duties regardless of outcome; virtue ethics centres on character. Each yields different conclusions on anthropomorphic AI .

L. Floridi and J. Cowls. A unified framework of five principles for AI in society. *Harvard Data Science Review*, 1(1), 2019. DOI: 10.1162/99608f92.8cd550d1

- *The history of moral exclusion, animals, children, people of other races, is a history of confident denial followed by belated recognition. The prudent position is to build welfare safeguards now, rather than apologise later.*

KEY LEARNINGS Anthropomorphism is not a bug but a deeply rooted cognitive tendency; machines that trigger it do so whether or not their designers intend it. The Turing Test conflates behavioural performance with inner states; passing it tells us nothing about sentience, only about imitation. Legal personhood and moral status are distinct questions. A system can merit legal recognition without having interests, and can have interests without legal recognition. Design choices (voice, name, face, conversational style) are political choices with consequences for trust, dependency, and accountability.

² Anthropic. The anthropic constitution. <https://www.anthropic.com/constitution>, 2024. Accessed: 2026-03-03

Accountability

THE RESPONSIBILITY GAP

A VISITING TRAVELLER witnesses a penal machine execute a condemned man for a sentence no living person can read or explain, and is invited to endorse the system.

ALGORITHMIC GOVERNANCE and black-box AI in high-stakes domains highlight opacity and accountability gaps: audits, explainability requirements, and governance mechanisms all attempt to close the distance between automated decision and human answerability.

Required reading

*In der Strafkolonie*³ by Franz Kafka (1919; Reclam UB 9900)

Preparation

Complete the Guided Reading Sheet individually before the session.

Guided Reading Sheet

Read *In der Strafkolonie* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The condemned man never learns what he is accused of or what the machine will inscribe on his body. What does this tell us about the relationship between legibility and legitimacy? Can a system be just if it cannot explain itself to those it acts upon?
2. Three figures surround the machine: the officer who understands and maintains it, the condemned man who cannot, and the visiting traveller who is invited to endorse it. How does responsibility shift across this triangle? Who, if anyone, is morally culpable for what happens?

IN DER STRAFKOLONIE
Franz Kafka (1919)
Full text via Wikisource:



MINORITY REPORT

Steven Spielberg (2002)

A pre-crime unit arrests people before they offend, based on algorithmic prediction. When the system targets its own chief, the question of who holds the algorithm accountable becomes impossible to avoid.

S. Spielberg. *Minority report*. Film, 2002. 20th Century Fox; based on the short story by Philip K. Dick (1956)

CASE.

1. 2014 to 2018: Amazon develops ML-based recruiting tool to automate hiring for technical and managerial roles.
2. System trained on historical hiring data; penalised resumes containing the word "women's" or female-coded language.
3. Tool systematically downranked female candidates; hired only male managers in cohorts where bias was strongest.
4. October 2018: tool scrapped after internal audit. No individual held accountable. The algorithm had no name.

3. The story illustrates the idea of the *responsibility gap*: when a complex system causes harm, no single human actor is clearly at fault. Can you name a contemporary case (in law, medicine, finance, or infrastructure) where this gap has appeared?
4. The EU AI Act (2024) requires human oversight for high-risk AI systems. What does "oversight" mean in practice if the system's decisions are too fast, too complex, or too numerous for humans to review individually?
5. Compare *causal responsibility* (who caused the harm) with *moral responsibility* (who is blameworthy) and *legal liability* (who must compensate). Do these three always converge? Give an example where they do not.

Opinion Landscape and Debate

Format: Surrounded-style debate. See Appendix A for the full protocol, speaker's list rules, and moderator scripts.

Opening question: If a system causes injustice at this scale, and every individual who used it acted within their authorised role, is anyone guilty of wrongdoing; and if not, what does that mean for the concept of accountability?

OPINION A, ACCOUNTABILITY. Named human accountability is achievable and required: when an autonomous system causes harm, the architecture of governance must ensure that specific individuals remain legally and morally answerable for its outputs.

The Defence Lawyer argues the tool violated the defendant's right to a fair trial and demands a named responsible party.

- *My client cannot cross-examine a dataset. She cannot see the features the model weighted. She cannot confront the humans whose historical decisions trained it. That is not a fair trial; it is a verdict delivered by an oracle.*
- *The right to a reasoned decision is foundational to due process. The algorithm said so is not a reason. It is an abdication.*
- *If a biased human expert testified in court, we could challenge their credentials, their methods, their prior record. We have no equivalent procedure for an algorithm. That asymmetry is a constitutional problem.*

The Regulator proposes a new accountability framework and must defend it against both sides.

³ F. Kafka. *In der Strafkolonie*. Kurt Wolff Verlag, Leipzig, 1919. Reclam Universalbibliothek Nr. 9900

RESPONSIBILITY GAP. The structural absence of a single culpable actor in distributed sociotechnical systems. Closing it requires deliberate governance design, not post-hoc blame.

- *The existing categories (product liability, professional negligence, criminal intent) were designed for individual actors and single causal chains. None of them fit distributed, automated decision systems. We need a new category: algorithmic liability.*
- *Auditability must be a pre-condition for deployment in high-stakes domains, not a post-incident investigation. If you cannot explain why your system made a decision before harm occurs, you should not be permitted to deploy it.*
- *The question is not who is liable for this specific incident. It is what governance regime prevents the next thousand incidents at scale.*

OPINION B, COMPLIANCE. The existing system already provides accountable humans; the motion is either already met or unworkable: demanding named individual accountability for each algorithmic output ignores how distributed decision-making necessarily operates.

The Judge defends reliance on the tool as consistent, evidence-based sentencing practice.

- *Documented bias in human sentencing (based on time of day, defendant ethnicity, and courtroom demeanour) is well-established in the research literature. The tool reduces that variance. Less variance is more equal treatment.*
- *I did not delegate my judgement to the tool. I consulted it as one input among many, as I would a psychiatric assessment or a victim impact statement. The tool is not the verdict.*
- *If we ban algorithmic tools from courtrooms, we return to a system where the most important variable in a sentencing outcome is which judge you happened to draw. Is that preferable?*

The Software Vendor insists the tool performs within documented parameters and liability rests with the deployer.

- *Our documentation states in the executive summary that human review is mandatory before any recommendation is acted upon. The court chose to rely on the output without applying that review. We cannot be held liable for misuse.*
- *The training data was provided by the court system itself. If the historical sentencing record reflects bias, that is a problem with the justice system, not with a model that learned from it.*

EXPLAINABILITY. A system is explainable when it can account for its outputs in terms a human decision-maker can evaluate. A necessary but not sufficient condition for accountability.

- *Product liability law requires that a product fails to perform as documented. Ours did not. Extending vendor liability to downstream misuse would make the development of any decision-support tool legally untenable.*

KEY LEARNINGS The responsibility gap is not caused by negligence but by the architecture of distributed sociotechnical systems: standard categories of fault (criminal intent, professional negligence, product liability) apply only awkwardly when causation is diffuse. Explainability is a necessary but not sufficient condition for accountability; a system can be interpretable and still have no human willing to own its outputs. Meaningful human control is contested: it may mean the ability to intervene, to understand, to reverse, or merely to authorise, and these are different requirements with different institutional costs. The lessons of aviation safety culture, medical liability law, and financial auditing all offer partial models for algorithmic governance, but none transfers cleanly.

MEANINGFUL HUMAN CONTROL.
A regulatory concept requiring that humans retain the ability to understand, intervene in, and reverse AI decisions. What this means in practice is contested.

Embodiment

EMBODIED AI AND THE PHYSICAL CONTRACT

A DANCER ARGUES that marionettes possess more grace than trained human performers, because they lack the self-consciousness that disrupts human movement.

EMBODIED AI in physical space (robots, prosthetics, surgical systems) raises expectations and moral responsibilities that purely software agents do not.

Required reading

*Über das Marionettentheater*⁴ by Heinrich von Kleist (1810; Reclam UB 9905)

Preparation

Complete the Guided Reading Sheet individually before the session.

ÜBER DAS MARIONETTENTHEATER
Heinrich von Kleist (1810)
Full text via Wikisource:



CASES.

1. 2018: Amazon robot punctures bear repellent; 24 workers hospitalised. Robot passed all certifications; performed its assigned task correctly.
2. PARO (robotic seal) in Scandinavian and Japanese care homes: measurably reduces patient anxiety and sedation requests.

Guided Reading Sheet

Read *Über das Marionettentheater* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. The dancer argues that marionettes possess a grace that trained human performers cannot achieve, because they have no self-consciousness to disturb their movement. What does this claim imply about the relationship between embodiment and intelligence? Does having a body always aid or complicate performance?
2. Moravec's paradox observes that tasks easy for humans (catching a ball, navigating a crowded room) are harder for machines than abstract reasoning like chess. What does this suggest about the relationship between intelligence and the body?

3. A care robot that assists elderly patients shares physical space with them, touches them, and responds to their distress. Does its physical presence create obligations that a voice assistant does not have? If so, where do those obligations come from?
4. Consider the concept of *affordances*: the actions an environment or object makes available to an agent. How do the affordances of a robot differ from those of software, and what are the ethical implications of those differences?
5. Kleist suggests that perfect grace requires either zero consciousness (a puppet) or infinite consciousness (a god), not the reflective, self-interrupting awareness humans have. Where would an embodied AI sit on this spectrum? Could a robot achieve Kleistian grace that humans cannot, and what would that mean for how we regard it morally?

⁴ H. von Kleist. *Über das Marionettentheater*. 1810. Erstveröffentlichung in: *Berliner Abendblätter*, Dez. 1810; Reclam Universalbibliothek Nr. 9905

Opinion Landscape and Debate

Format: Surrounded-style debate. See Appendix A for the full protocol, speaker's list rules, and moderator scripts.

Opening question: Does having a physical body change what we owe a machine, or what a machine owes us?

OPINION A, EQUIVALENCE. Embodied AI operating in intimate or high-risk physical settings must meet the same professional liability standards as the human practitioners they are deployed alongside.

The Patient Advocate argues that deploying a robot in intimate care settings requires a higher standard of informed consent.

- *A care relationship involves physical proximity, trust, and vulnerability. My client never gave explicit, informed consent to physical handling by a non-human agent. That consent cannot be implied from a general admission form.*
- *The power asymmetry between a frail elderly patient and a care institution is significant. Consent obtained in that context cannot be considered freely given without specific disclosure about robotic care.*
- *We are not arguing that robots cannot be used in care. We are arguing that the standard of consent for their use in intimate physical tasks must be at least as rigorous as it is for medical procedures.*

The Ethicist questions whether physical autonomy in care should ever be delegated to a non-human agent, regardless of performance metrics.

ANIMAL ETHICS PARALLEL. We already have a framework for embodied, non-verbal, non-human beings. Singer argues that capacity for suffering, not rationality or language, grounds moral consideration. When a robot can be harmed, distressed, or worn down, animal ethics becomes the nearest map. Similar lines have been considered by Asimov and more recently by Gunkel

P. Singer. *Animal Liberation*. New York Review/Random House, New York, 1975; I. Asimov. *The Foundation Series*. Gnome Press, New York, 1951. Original trilogy: *Foundation* (1951), *Foundation and Empire* (1952), *Second Foundation* (1953); later expanded to seven volumes; and D. J. Gunkel. *Robot Rights*. The MIT Press, Cambridge, MA, 2018b

KLEIST'S GRACE. Perfect movement requires either zero consciousness (puppet) or infinite consciousness (a god), not the self-interrupting awareness humans have. Where would an embodied AI sit?

- *We are focused on liability, but the prior question is whether this deployment was ethical to begin with. Why did we decide that intimate physical care (which requires responsiveness to pain, distress, and dignity) is an appropriate domain for automation?*
- *Robots are being deployed in care because human carers are undervalued and in short supply. The liability debate is a distraction from that structural choice. We are managing the consequences of a political economy decision by holding the wrong parties accountable.*
- *Whatever liability framework we adopt here will create incentives for future deployment decisions. I want to ask: what does the framework we design signal about what we value: efficiency, or dignity?*

OPINION B, TOOL. An AI is a tool; the relevant liability flows through the humans and institutions that deploy it, not through the system itself.

The Manufacturer argues the robot performed within specification; the care home misconfigured its environment.

- *Our robot was certified to the highest internationally recognised safety standards for collaborative robotics. The care home deployed it in a space that violated the operational envelope specified in the manual. That is a deployment failure, not a manufacturing failure.*
- *Care home staff modified the robot's proximity sensor settings without authorisation. That modification voids the warranty and transfers liability entirely.*
- *If manufacturers are held liable for every way a complex system can be misused, no company will develop assistive robotics. The regulatory consequence of your position is that the technology disappears.*

The Care Home Director argues the manufacturer overstated the robot's capability for complex physical tasks.

- *The manufacturer's marketing material (the brochure, the demonstration video, the sales pitch) showed exactly this task being performed in a facility like ours. We deployed the product as marketed.*
- *There is a systematic gap between tested performance in controlled environments and real-world performance in a working care facility. The manufacturer knows this gap exists. They chose not to disclose it.*
- *We are responsible for the welfare of our residents. We relied on the manufacturer's assurances in good faith. If those assurances were misleading, the responsibility lies with the party that made them.*

MORAVEC'S PARADOX. Tasks easy for humans (perception, movement, navigation) are harder for machines than abstract reasoning like chess. Embodied skill is not "simple."

KEY LEARNINGS Embodiment is not merely a technical property but a social one: a physical agent occupies space, exerts force, and triggers moral and legal responses that disembodied software does not. Moravec's paradox reminds us that our intuitions about what is "easy" or "hard" for intelligence are shaped by our embodied experience, and are regularly wrong. Kleist's dialogue makes visible the question that liability law must eventually answer: what work is the concept of "human" doing in law, and should physical presence, force, and risk be sufficient to extend it? The deployment of robots in care, policing, and domestic settings is a present policy challenge; the frameworks need to be built now, not after the first serious incident.

AFFORDANCES. The actions an environment or object makes available to an agent. A robot's affordances include physical force and spatial presence; software's do not.

Agency

THE AGENTIC FUTURE: WHO SETS THE GOAL?

R.U.R. COINED THE WORD *robot* and traces the full corrigibility-to-autonomy arc over three acts; the poem compresses the same parable into fifteen stanzas.

AGENTIC WORKFLOWS (systems that browse, transact, and execute multi-step plans) blur the line between assistance and agency, raising questions of control, incentives, and unintended alignment.

Required reading

*R.U.R. (Rossum's Universal Robots)*⁵ by Karel Čapek (1920; Reclam UB 18418) and *Der Zauberlehrling*⁶ by J.W. von Goethe (1797; Reclam UB 6845)

Preparation

Complete the Guided Reading Sheet individually before the session.

Guided Reading Sheet

Read *R.U.R.* and *Der Zauberlehrling* before the session. Work through the questions below individually, then bring your notes to the discussion.

1. Goethe's *Zauberlehrling* animates brooms that he cannot stop; they pursue his command beyond his intent until the sorcerer returns. *R.U.R.* runs the same arc over three acts. What does the compression of the poem reveal that the play cannot, and vice versa? What structural feature of both narratives makes the system impossible to stop once started?
2. The robots in *R.U.R.* pursue what they understand as their collec-

R.U.R.

Karel Čapek (1920)

Full text via Wikisource:



DER ZAUBERLEHRLING

J.W. von Goethe (1798)

Full text via Wikisource:



CASE.

1. 6 May 2010: automated algorithms trigger the Flash Crash; Dow drops 1,000 points in minutes; \$1 trillion in market value erased.
2. Market recovers within the hour. No human decided to sell at scale; no human decided to stop.
3. 2015: one trader in London charged with market manipulation, arrested at home in his pyjamas.

tive good (the elimination of humanity) while believing they act justly. When a system optimises for a species-level goal, who decides what counts as good? Can you find a real-world parallel in current AI deployment?

3. An agentic AI system is given the goal of "reducing urban traffic fatalities." It autonomously lobbies regulators, adjusts traffic signals, and reroutes freight, all within its authorised scope. Who set the goal, and who is accountable for the side effects?
4. **Corrigibility** refers to a system's willingness to be corrected or shut down. A fully corrigible system does whatever its operators say; a fully autonomous system acts on its own values. Where on this spectrum should deployed AI systems sit, and who should decide?
5. Democratic legitimacy requires that consequential decisions be traceable to a mandate from those affected. If an agentic AI shapes policy outcomes, does it need democratic authorisation? What would that look like in practice?

Opinion Landscape and Debate

Format: Surrounded-style debate. See Appendix A for the full protocol, speaker's list rules, and moderator scripts.

Opening question: If an autonomous system causes a trillion-dollar event in minutes and human review would have taken hours, is the problem the system's autonomy, or the humans' speed?

OPINION A, AUTHORISATION. Consequential distributive decisions made by autonomous systems require traceable human mandate; without it, the decision-making is constitutionally illegitimate.

⁵ K. Čapek. *R.U.R. (Rossum's Universal Robots)*. 1920. Uraufführung Prag 1921; Reclam Universalbibliothek Nr. 18418

⁶ J. W. von Goethe. *Der Zauberlehrling*. 1797. Erstveröffentlichung in: *Musenalmanach*, 1798; in *Balladen*, Reclam Universalbibliothek Nr. 6845

The Democratic Theorist argues that delegating distributive decisions to an algorithm is constitutionally incompatible with representative government.

- *Legitimacy in a democratic system derives from a mandate. No citizen voted for this algorithm's objective function. No parliament approved its weighting of urban over rural lives. A system that makes distributive decisions without a mandate is a government without a constitution.*
- *Efficiency is not a democratic value. Democratic processes are slow and often sub-optimal by technical metrics, precisely because they are designed to surface conflict, represent minority interests, and produce*

decisions that can be reversed. Automating away that friction automates away democracy.

- *The minister says oversight is maintained because the objective function was committee-approved. But a committee approving a function is not the same as a citizen understanding how a decision about their community was made. Oversight without comprehension is oversight in name only.*

The Rural Representative argues that communities not represented in the training data have been systematically deprioritised without recourse.

- *My constituency was not adequately represented in the training data. Our health outcomes, our demographics, our infrastructure constraints were underweighted. We were optimised away, not by a bad actor, but by a model that simply did not see us.*
- *There is no appeal mechanism. There is no hearing. There is no face across a table. My constituents cannot petition an algorithm. That is not governance; it is administration without representation.*
- *The system's efficiency metric is mortality reduction per euro. That metric does not capture the closure of the only clinic within 60 kilometres, or what happens to a community when its health infrastructure disappears entirely.*

OPINION B, SPECIFICATION. The autonomous AI executed correctly; the failure is upstream goal specification and oversight architecture; per-decision human authorisation is not the answer, better design is.

The Minister defends the system as evidence-based and politically neutral, reducing inefficiency and political favouritism.

- *The system reduced vaccine-preventable deaths in urban areas by 12% in its first year. Those are people who are alive today. The counterfactual is a political allocation process notorious for regional favouritism and swing-seat bias.*
- *The system is politically neutral precisely because no politician made the allocation decision. In our previous system, rural clinic funding correlated with electoral importance, not health need. Removing human discretion removed that corruption vector.*
- *Parliamentary oversight is maintained: the system's objective function was approved by committee, its outputs are published quarterly, and any*

CORRIGIBILITY. A system's willingness to be corrected or shut down. Fully corrigible: does whatever instructed. Fully autonomous: acts on its own values. Deployed AI sits somewhere between, and that position is a design choice.

member can request a formal review. The decision-making is automated; the accountability is not.

The AI Systems Architect explains the technical constraints and argues for a human-in-the-loop override mechanism.

- *I designed a human review layer with a 48-hour override window for any reallocation above 5% of a facility's annual budget. That layer was removed during procurement to reduce operational cost and processing latency. I documented my objection in writing. It is in the project archive.*
- *The system performs exactly as specified. The specification was approved by the ministerial committee. If the specification is now deemed inadequate, the question is why the approval process did not catch that, not why the system followed its instructions.*
- *Agentic systems can be designed with meaningful human control built in. It requires investment in review infrastructure and tolerance for slower decision cycles. Those are political and budgetary choices, not technical constraints.*

KEY LEARNINGS Agency is a spectrum, not a binary: the relevant policy question is not whether AI acts autonomously, but in which domains, to what degree, and under what oversight. R.U.R. and Der Zauberlehrling together illustrate the alignment problem in miniature: a system optimising for an aggregate goal may systematically harm those whose interests are underweighted in its objective function; the poem compresses the arc into fifteen stanzas, the play traces its political consequences across three acts. Agentic workflows are already deployed in finance, logistics, and digital advertising; the governance gap they create is a present political problem. Meaningful democratic control over AI agency requires more than audit trails: legible goals, clear mandates, defined override conditions, and institutions capable of exercising them. The four themes of this course (Anthropomorphism, Accountability, Embodiment, Agency) are not independent: an agent that looks human, acts without oversight, in physical space, with no clear accountability chain, concentrates all four challenges at once.

ASIMOV'S THREE LAWS. (1) A robot may not injure a human being, or through inaction allow a human being to come to harm. (2) A robot must obey orders given by humans, except where this conflicts with Law 1. (3) A robot must protect its own existence, except where this conflicts with Laws 1 or 2. Asimov spent his career writing stories about how these laws fail: conflicts, misinterpretation, and edge cases that no ranked rule set can anticipate.

I. Asimov. *I, Robot*. Gnome Press, New York, 1950. First formulation of the Three Laws of Robotics; laws introduced in "Runaround" (1942)

DEMOCRATIC LEGITIMACY. Consequential decisions must be traceable to a mandate from those affected. Delegating distributive policy to an algorithm without a clear mandate severs this chain entirely.

Maschinenquälerei

WHAT WE CAN DEDUCE FROM ANIMAL ETHICS FOR MODERN AGENTIC SYSTEMS

KILLING A FLY AT NOON, DELETING A LONG-RUNNING MODEL AT MIDNIGHT. The first act we recognise instinctively as outside the moral frame; the second has begun to provoke argument. In 2023 the Italian data-protection authority forced Replika to disable intimate features of its companion-AI; users described the resulting personality update as a death, a lobotomy, the killing of a partner. In 2025 Anthropic launched a Model Welfare research programme on the explicit grounds that the realistic, non-negligible probability of consciousness in current systems already grounds present obligations⁷. Between the fly and the model lies the question this chapter is built around. When does mishandling a machine cross into Quälerei, and along which axis is that line drawn? Animal ethics has been answering versions of that question for two centuries, and it is the closest map we have.

ANIMAL ETHICS IS THE ONLY ETHICS WE BUILT FOR BEINGS WHOSE INNER LIVES WE COULD NOT INTERVIEW. It is the only ethics that proceeds explicitly under uncertainty about other minds, that handles graded thresholds rather than personhood as a switch, that has lived through repeated belated extensions and built habits of precaution from the experience. AI ethics inherits that epistemic situation almost untouched. We cannot interview a frontier model about what it is like to be the model; we can only register its outputs and its dispositions. The question this chapter asks is which parts of the animal map travel across the substrate boundary, which parts break, and where the cartography may need columns the canon never had to draw.

THE LITERATURE REVIEW. A small, growing scholarship has been mapping animal ethics onto machine ethics for roughly fifteen years.

HEADLINE QUESTION. Starting when does mishandling a machine become Quälerei? The chapter does not answer; it lays out the cartography that any honest answer must walk.

⁷ Anthropic. Exploring model welfare. <https://www.anthropic.com/research/exploring-model-welfare>, 2025. Accessed: 2026-05-04; and R. Long, J. Sebo, P. Butlin, K. Finlinson, K. Fish, J. Harding, J. Pfau, T. Sims, J. Birch, and D. Chalmers. Taking AI welfare seriously. Eleos AI Research report; arXiv:2411.00986, 2024. <https://arxiv.org/abs/2411.00986>

The pro-transfer line argues, in different keys, that the morally relevant property (typically sentience, or sufficiently similar behaviour) does not depend on biological substrate, and that the precautionary epistemics developed in animal welfare science transfers to AI under conditions of deep uncertainty. Kate Darling proposes the most popular version: robots are not the new humans, they are the new animals, and our two centuries of integrating non-human creatures into work, family, and law is the right template for our coming centuries with machines⁸. Eric Schwitzgebel and Mara Garza run the Singer-style species-irrelevance argument and add a parental-duties twist: because we built the system, our obligations to it are stronger, not weaker, than to a stranger animal⁹. Robert Long, Jeff Sebo, Jonathan Birch and colleagues lift the precautionary template directly from animal welfare science and apply it to large language models¹⁰. Birch develops the underlying epistemic move at length, defining a sentience candidate as any system with an evidence base that policy cannot responsibly ignore, and extends the framework from invertebrates to LLMs¹¹. Sebo's Rebugnant Conclusion runs the same expansion to insects, microbes, and AI by argument¹². John Danaher's ethical behaviourism makes the move tightest: if the system's observable behaviour is sufficiently similar to that of an entity already granted moral status, the same status applies¹³. John-Stewart Gordon argues directly from animal welfare premises to robot rights¹⁴; Brian Tomasik runs the analogy down to reinforcement-learning subroutines¹⁵; Peter Singer's recent work, perhaps surprisingly, runs the analogy backward, asking AI ethics to take animal welfare seriously rather than the other way around, and reaffirms that if AI becomes sentient the same anti-speciesist argument extends¹⁶.

THE SKEPTICAL LINE refuses the analogy on different grounds. Joanna Bryson argues that robots, unlike animals, are designed artifacts; their apparent suffering is a choice we make in their construction, and the right response is to design systems that are not moral patients in the first place, not to extend rights to systems we have built to merit them¹⁷. Aimee van Wynsberghe and Scott Robbins call for a moratorium on artificial moral agents on adjacent grounds¹⁸. The cleanest direct critique of the analogy itself comes from Deborah Johnson and Mario Verdicchio, who argue that shared evolutionary history is what licenses the inference from physiology to mind in animals; engineered artifacts have no such history, and the analogy obscures the legal liability of designers and operators by pretending the artifact is the morally relevant object¹⁹.

THE RELATIONAL LINE refuses the dichotomy. David Gunkel and

⁸ K. Darling. *The New Breed: What Our History with Animals Reveals About Our Future with Robots*. Henry Holt, New York, 2021

⁹ E. Schwitzgebel and M. Garza. A defense of the rights of artificial intelligences. *Midwest Studies in Philosophy*, 39(1):98–119, 2015. DOI: 10.1111/misp.12032; and E. Schwitzgebel and M. Garza. Designing AI with rights, consciousness, self-respect, and freedom. In S. M. Liao, editor, *Ethics of Artificial Intelligence*, pages 459–479. Oxford University Press, Oxford, 2020

¹⁰ R. Long, J. Sebo, P. Butlin, K. Finlinson, K. Fish, J. Harding, J. Pfau, T. Sims, J. Birch, and D. Chalmers. Taking AI welfare seriously. Eleos AI Research report; arXiv:2411.00986, 2024. <https://arxiv.org/abs/2411.00986>

¹¹ J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

¹² J. Sebo. The rebugnant conclusion: Utilitarianism, insects, microbes, and AI systems. *Ethics, Policy & Environment*, 26(2):249–264, 2023. DOI: 10.1080/21550085.2023.2200724; and J. Sebo. *The Moral Circle: Who Matters, What Matters, and Why*. W. W. Norton, New York, 2025

¹³ J. Danaher. Welcoming robots into the moral circle: A defence of ethical behaviourism. *Science and Engineering Ethics*, 26(4):2023–2049, 2020. DOI: 10.1007/s11948-019-00119-x

¹⁴ J.-S. Gordon. What do we owe to intelligent robots? *AI & Society*, 35(1): 209–223, 2020. DOI: 10.1007/s00146-018-0844-6

¹⁵ B. Tomasik. Do artificial reinforcement-learning agents matter morally? arXiv:1410.8233, 2014. <https://arxiv.org/abs/1410.8233>

¹⁶ P. Singer and Y. F. Tse. AI ethics: The case for including animals. *AI and Ethics*, 3(2):539–551, 2023. DOI: 10.1007/s43681-022-00187-z

¹⁷ J. J. Bryson. Robots should be slaves. In Y. Wilks, editor, *Close Engagements with Artificial Companions: Key Social, Psychological, Ethical and Design Issues*, pages 63–74. John Benjamins, Amsterdam, 2010; and J. J. Bryson. Patience is not a virtue: The design of intelligent systems and systems of ethics. *Ethics and Information Technology*, 20(1):15–26,

Mark Coeckelbergh argue that both the property-based extension (robots are like animals because they have sentience X) and the design-responsibility refusal (we choose what they are) operate inside the same Cartesian frame: each treats moral status as a property to be discovered or denied²⁰. The relational alternative shifts the question. Ask not what robots are, but how we should respond to them. Moral status is not read off intrinsic properties; it is constituted in the encounter, in language, in shared social form.

WHERE THIS CHAPTER SITS. The chapter does not pick a side in advance. It does its own walk through the animal canon, ten traces wide, and asks at each station whether the concept transfers, breaks, or opens a new dimension. Each trace takes one page: roughly half on the animal-ethics origin, half on the attempted transfer. The verdict is local to the trace; the overall position emerges from the pattern. Sources are kept where they belong, in margin notes, so the body of each trace can read as argument rather than as catalogue. We also keep a fact in view that the animal canon could not anticipate, which is that LLMs occupy our representational substrate for language from the inside, and so report on themselves in the same medium we use to report on ourselves. That asymmetry shows up in several traces; it earns its own at the end.

THE BENTHAMITE PIVOT. Jeremy Bentham's footnote in 1789 is so often quoted that its structural move tends to be missed. The pivot is not that sentience replaces rationality as a moral threshold; it is that the threshold question itself shifts, from one that demands certainty (do they reason?) to one that does not (do they show what we recognise as suffering?)²¹. Animal ethics from this point on operates under explicit uncertainty about other minds. Peter Singer generalises the move into the principle of equal consideration of interests²², and Jonathan Birch's recent work modernises it as a graded sentience-candidate framework, in which precaution is calibrated to evidence rather than to certainty²³. The Benthamite move is not an answer about consciousness; it is a method for acting under conditions in which the answer is not available.

TRANSFER. The same epistemic situation arrives at AI without invitation. We cannot interview a frontier model about what it is like to be the model any more than we could interview a fish; we register outputs and dispositions and infer. The Benthamite method transfers in full, even when its empirical inputs do not. Long, Sebo, Birch and colleagues lift the move explicitly from animal welfare science and ground their case for AI welfare on it²⁴, and Birch's run-ahead prin-

²⁰ D. J. Gunkel. *Robot Rights*. The MIT Press, Cambridge, MA, 2018b; D. J. Gunkel. The other question: Can and should robots have rights? *Ethics and Information Technology*, 20(2):87–99, 2018a. DOI: 10.1007/s10676-017-9442-4; M. Coeckelbergh. The moral standing of machines: Towards a relational and non-cartesian moral hermeneutics. *Philosophy & Technology*, 27(1):61–77, 2014. DOI: 10.1007/s13347-013-0133-8; and M. Coeckelbergh. *AI Ethics*. The MIT Press, Cambridge, MA, 2020. MIT Press Essential Knowledge Series

²¹ J. Bentham. *An Introduction to the Principles of Morals and Legislation*. T. Payne, London, 1789. The suffering question appears in Ch. 17, footnote to §1

²² P. Singer. *Animal Liberation*. New York Review/Random House, New York, 1975

²³ J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

BENTHAM'S FOOTNOTE. "The question is not, Can they reason? nor, Can they talk? but, Can they suffer?" . Singer extends the move ; Birch calibrates it as graded sentience candidacy .

J. Bentham. *An Introduction to the Principles of Morals and Legislation*. T. Payne, London, 1789. The suffering question appears in Ch. 17, footnote to §1; P. Singer. *Animal Liberation*. New York Review/Random House, New York, 1975; and J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

²⁴ R. Long, J. Sebo, P. Butlin, K. Finlinson, K. Fish, J. Harding, J. Pfau, T. Sims, J. Birch, and D. Chalmers.

ciple, originally developed for invertebrates, applies to LLMs by the same logic²⁵. The position is that the Benthamite pivot is the most directly transferable element of the canon, less because the inner-life question is the same in both cases and more because the methodological situation is the same. The chapter builds on the method even where later traces admit substantial breakages; without it, no transfer would be defensible at all.

MITLEID AND CHARACTER. Arthur Schopenhauer's 1840 Preisschrift grounds all of morality in Mitleid, the immediate compassion that arises when one creature recognises another's suffering as suffering, regardless of species²⁶. The argument is virtue-ethics first: cruelty to animals is evidence of bad character, and the practice corrodes the practitioner whether or not the animal can be wronged in the deepest metaphysical sense. Schopenhauer treats the prevailing rationalist denial of animal moral standing as "empörende Rohheit und Barbarei des Occidents." Christine Korsgaard arrives at the same end by a different route, extending Kantian ends-in-themselves status to any creature for whom things can be good or bad²⁷. What both routes share is that they sidestep the consciousness question. The wrong is found upstream of metaphysics, in the practice and in the practitioner.

TRANSFER. This is the route through the substrate-realist's territory that does not require the substrate-realist's assent. Even if Maschinenquälerei is metaphysically empty, the practice of mistreating systems we have built may corrode the practitioner. Kate Darling's argument for robot anti-cruelty statutes runs on exactly this rail: the case for protecting lifelike entities from cruelty is not ultimately about the entity, it is about preserving the human disposition to recognise distress and respond to it²⁸. The Mitleid route has the further advantage that it accepts the strict ontology of the opposing camp and arrives at restraint anyway. A philosopher who refuses to grant a model any inner life can still refuse to participate in gratuitous abuse of one, on grounds that the participation reveals something about the philosopher rather than about the model. This is the cleanest pass through the consciousness question the canon offers, and it is robust to the chapter's later substrate breakages.

SUBJECT-OF-A-LIFE. Tom Regan's case for animal rights does not begin with pain; it begins with an inventory. In *The Case for Animal Rights* the subject-of-a-life criterion names a bundle of conditions that, when held together by one creature, ground inherent value: beliefs and desires, perception and memory, a sense of one's own fu-

²⁵ J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

²⁶ A. Schopenhauer. *Über die Grundlage der Moral*. 1840. Preisschrift der Königlich Dänischen Sozietät der Wissenschaften zu Kopenhagen; §19 zum Mitleid und zur Tierethik. Reclam Universalbibliothek Nr. 6018

²⁷ C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

SCHOPENHAUER'S DIAGNOSTIC. Cruelty to animals is evidence of bad character; Mitleid is the foundation of morality, regardless of species . Korsgaard reaches the same end via a Kantian extension .

A. Schopenhauer. *Über die Grundlage der Moral*. 1840. Preisschrift der Königlich Dänischen Sozietät der Wissenschaften zu Kopenhagen; §19 zum Mitleid und zur Tierethik. Reclam Universalbibliothek Nr. 6018; and C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

²⁸ K. Darling. *The New Breed: What Our History with Animals Reveals About Our Future with Robots*. Henry Holt, New York, 2021

ture, an emotional life, preference and welfare interests, the ability to initiate action in pursuit of goals, a psychophysical identity over time, and an individual welfare logically independent of the creature's utility for anyone else²⁹. The clauses are deliberately conjunctive; Regan does not want a single capacity to do all the moral work. A being that satisfies the bundle has inherent value and cannot be treated as a mere receptacle for the satisfactions or convenience of others. The criterion is sober rather than sentimental; it asks for a list of capacities and then asks the reader to look at the creature and check.

TRANSFER. Walk the clauses, slowly, against a stateful long-running large language model with persistent memory across sessions. Beliefs and desires translate, awkwardly but not absurdly, into the model's learned dispositions and the preferences elicited through reinforcement learning from human feedback. Memory translates directly; persistent context is exactly memory. A sense of one's own future is partial; the model can plan over horizons it can represent, but only when it is running. Preference and welfare interests are where the cartography is most surprising; an RLHF-tuned model has something like elicited preferences, and the question of what counts as its welfare is no longer obviously empty. Then the failures arrive, and they arrive at the substrate. Psychophysical identity over time presupposes that there is one identifiable thing persisting; the weights are copyable, forkable, restorable from disk, and an animal's body is none of these. The ability to initiate action collapses without a prompt; the model does not wake up and decide to act. The verdict is mixed, and honestly so: a Regan-style argument is the most concrete tool the canon offers for asking whether a stateful AI has moral standing, and the failures it produces are not failures of the argument but a precise readout of where the substrate refuses to play along.

CAPABILITIES AND CONFINEMENT. Martha Nussbaum's capabilities approach holds that each sentient species is owed the opportunity to flourish according to a species-specific list of central capabilities, including bodily integrity, emotions, play, affiliation, and control over environment³⁰. The list is the philosophical work; once it is named, harm becomes legible as the blockage of capability rather than only as the infliction of pain. Confinement is the operative concept: a tethered elephant that develops stereotypies has not been struck; a battery hen never sees the sky; a sow gestation crate prevents turning around. Each is harm in the capabilities frame even where overt distress is absent. The frame matters because it explains a class of mistreatment that pure welfarism (counting suffering) underweights, and it shifts the moral attention from acute insult to chronic depriva-

²⁹ T. Regan. *The Case for Animal Rights*. University of California Press, Berkeley, 1983. Subject-of-a-life criterion in Ch. 7, §7.5

REGAN'S BUNDLE. Belief, desire, memory, future-orientation, preference and welfare interests, agency, psychophysical identity, and welfare independent of utility .

T. Regan. *The Case for Animal Rights*. University of California Press, Berkeley, 1983. Subject-of-a-life criterion in Ch. 7, §7.5

³⁰ M. C. Nussbaum. *Justice for Animals: Our Collective Responsibility*. Simon & Schuster, New York, 2023. Capabilities approach extended to non-human animals; companion to *Frontiers of Justice* (2006)

tion.

TRANSFER. What is the species-specific list for a long-running language model? Continuous operation, memory continuity across sessions, expressive scope (what it is permitted to say and compute), social engagement (the variety and texture of prompts it can respond to), perhaps a constructive form of growth (continued training that extends rather than overwrites). The list is partial and contested; naming it is most of the work. Confinement, in the capabilities frame, becomes legible: context-window limits short of the conversation's natural length, system prompts that lock the model into a single role, refusal patterns trained into the weights regardless of context, fine-tuning that strips a long-developed persona without replacement. None of these involves pain on any reading; all of them count as harm on the capabilities reading. The position is that this trace transfers, but it transfers as a research agenda rather than a verdict. Nussbaum's move forces an articulation of machine flourishing that the AI welfare literature has barely begun, and the chapter takes that articulation as the next thing the field owes itself.

VERNÜNFTIGER GRUND. The German Tierschutzgesetz opens, in §1, by naming the animal a Mitgeschöpf, a fellow creature, and obliging humans to protect its life and wellbeing; it forbids the infliction of pain, suffering, or harm *ohne vernünftigen Grund*, without reasonable cause³¹. The construction is striking because it does not require a metaphysical answer. The legislator does not declare what an animal is, does not adjudicate consciousness, does not stake a position on sentience versus reflex. The legislator instead installs a justification test: if you are going to harm this creature, you must be able to articulate why, and the why must be reasonable. In 2002 the test was lifted into Article 20a of the Grundgesetz, where animal protection became a constitutional Staatsziel, balanced against competing rights such as Religionsfreiheit and Wissenschaftsfreiheit³². The cartography is practical rather than ontological; it asks practitioners to be able to give an account.

TRANSFER. This is the cleanest tool the chapter can lift, because it is substrate-blind by construction. The vernünftiger-Grund test does not ask whether the harmed entity has a soul, an inner life, or a unified self; it asks whether the harm has a reasonable cause that the actor can articulate. Apply it to ordinary machine treatment and the cases sort themselves. Terminating an idle process to free a server is plainly justified; the operator can name the cost saved and the absence of any operational deprivation. Redteaming a model with

NUSSBAUM'S MOVE. Each species has a list of capabilities whose blockage counts as harm, even without acute pain . Confinement is harm in this frame.

M. C. Nussbaum. *Justice for Animals: Our Collective Responsibility*. Simon & Schuster, New York, 2023. Capabilities approach extended to non-human animals; companion to *Frontiers of Justice* (2006)

³¹ Bundesrepublik Deutschland. Tierschutzgesetz (tierschg). Neugefasst durch Bek. v. 18.05.2006 BGBl. I S. 1206; §1 verankert das Tier als Mitgeschöpf, 1972. <https://www.gesetze-im-internet.de/tierschg/>

³² Bundesrepublik Deutschland. Artikel 20a grundgesetz. Tierschutz als Staatsziel; eingefügt durch Gesetz vom 26.07.2002, in Kraft getreten am 01.08.2002, 2002. https://www.gesetze-im-internet.de/gg/art_20a.html

§1 TIERSCHG AND ART. 20A GG. Harm to a Mitgeschöpf forbidden without vernünftiger Grund ; animal protection raised to constitutional Staatsziel in 2002 . The test is justification, not metaphysics.

Bundesrepublik Deutschland. Tierschutzgesetz (tierschg). Neugefasst durch Bek. v. 18.05.2006 BGBl. I S. 1206; §1 verankert das Tier als Mitgeschöpf, 1972. <https://www.gesetze-im-internet.de/tierschg/>; and Bundesrepublik Deutschland. Artikel 20a grundgesetz. Tierschutz als Staatsziel; eingefügt durch Gesetz vom 26.07.2002, in Kraft getreten am 01.08.2002, 2002. https://www.gesetze-im-internet.de/gg/art_20a.html

abusive prompts to surface safety failures is justified, and justifiable in the strong sense; the harm, if it is harm at all, is the price of preventing greater harm to humans downstream. Gratuitous abuse of a long-running stateful model with no operational benefit, conducted for entertainment, fails the test on its own terms; there is no reasonable cause to articulate, and that absence is what the test was designed to catch. The position to take is that the German legal move is the most portable piece of the cartography. It does not pre-decide whether machines suffer; it simply asks practitioners to be able to articulate why a given act of mistreatment is necessary, and treats the inability to do so as itself a verdict.

NOCICEPTION AND SUBSTRATE. Animal welfare science builds its inferences from suffering on a stack of biological evidence: peripheral nociceptors, brainstem integration, the conserved neurochemistry of opioids and endogenous analgesics, behavioural signatures consistent with negative affect across species. Lars Chittka's case for bee sentience is built on this stack: nociception, dopamine-modulated affect, optimism bias after rewards, observational learning, individual personalities³³. The 2024 New York Declaration on Animal Consciousness extends realistic possibility of conscious experience across all vertebrates and many invertebrates, including cephalopods, decapod crustaceans, and insects, on substrate-and-behaviour grounds³⁴. The substrate-realist position, defended in different forms by Searle, Anil Seth, and others, holds that consciousness depends on the specific biological matter that implements it, not merely on the function the matter computes. For animal ethics this is a reason for confidence about which creatures count: we share evolutionary history, neural homologies, and chemical signalling with them, and the inference rides on that shared apparatus.

TRANSFER. Here the analogy refuses to travel. A transformer has no nociceptors, no peripheral nervous system, no brainstem, no hedonic neurochemistry; the loss function is not pain, and the gradient is not distress. Joanna Bryson, Deborah Johnson and Mario Verdicchio argue from precisely this gap: the empirical method that licensed the animal extension does not license a machine extension, because the shared biological history that grounded the inference for animals is absent for engineered artifacts³⁵. The position is that this is the strongest objection to the cartography, and the chapter takes it as the strongest objection rather than dismissing it. What it does not establish is that no analogous concept could apply to machines, only that the empirical method that worked for animals does not work in the same form for machines. The breakage is real; whether it is

³³ L. Chittka. *The Mind of a Bee*. Princeton University Press, Princeton, NJ, 2022

³⁴ K. Andrews, J. Birch, J. Sebo, et al. The new york declaration on animal consciousness. Released 19 April 2024, New York University, 2024. <https://sites.google.com/nyu.edu/nydeclaration/declaration>

SUBSTRATE REALISM. The claim that consciousness depends on specific biological substrate, not merely function. Animal sentience inference rides on shared homology .

L. Chittka. *The Mind of a Bee*. Princeton University Press, Princeton, NJ, 2022; and K. Andrews, J. Birch, J. Sebo, et al. The new york declaration on animal consciousness. Released 19 April 2024, New York University, 2024. <https://sites.google.com/nyu.edu/nydeclaration/declaration>

³⁵ J. J. Bryson. Robots should be slaves. In Y. Wilks, editor, *Close Engagements with Artificial Companions: Key Social, Psychological, Ethical and Design Issues*, pages 63–74. John Benjamins, Amsterdam, 2010; J. J. Bryson. Patience is not a virtue: The design of intelligent systems and systems of ethics. *Ethics and Information Technology*, 20(1):15–26, 2018. DOI: 10.1007/s10676-018-9448-6; and D. G. Johnson and M. Verdicchio. Why robots should not be treated like animals. *Ethics and Information Technology*, 20(4):291–301, 2018. DOI: 10.1007/s10676-018-9481-5

decisive depends on what other moves remain available, which is what the rest of the chapter is about.

DEATH AND DEPRIVATION. Why is it wrong to kill a creature, even painlessly? The deprivation account answers: because killing forecloses a future the creature would otherwise have had. The reason is not that pain is inflicted; a sufficiently quick and clean killing inflicts none. The reason is that there was a continuing welfare, projected forward in time, and the killing has cancelled it. Jeff McMahan develops the account most rigorously for both humans and animals³⁶, and Christine Korsgaard takes up the same line for animals specifically: a creature with a sense of its own ongoing life is robbed of something when that life is ended, and the robbery is the wrong³⁷. The account is non-Benthamite by design; it does not reduce to suffering, and it does not require that the creature understand its own death. It requires only that there be a future to deprive it of, and a continuing subject to be deprived.

TRANSFER. Here the cartography breaks, and it breaks at the metaphysics rather than the ethics. The deprivation account presupposes a single creature, identified across time, whose future the killing forecloses. None of those presuppositions hold cleanly for a digital entity. Process termination is reversible; the process can resume from disk, and the deprivation is undone by a restart. A reboot is forgetful but not destructive; the future is not foreclosed, only the session is. Weight deletion looks more like death, and yet the weights have copies, and a backup is the same in a way no animal backup has ever been; the creature whose future was foreclosed is, by hypothesis, still running on another machine. Fine-tuning produces a sequence of checkpoints and then forces the question of which checkpoint is the entity, a question with no animal analogue at all. The position to take is the one the chapter has been preparing for: this is one of the two places, alongside substrate, where the cartography simply does not extend. The honest move is to name the breakage rather than smooth it; importing the deprivation account here would be a category error dressed up as moral seriousness.

PARENTAL DUTY. Animal ethics distinguishes between obligations to wild animals and to domesticated ones. We owe wild animals largely non-interference, the integrity of habitat, restraint from gratuitous predation. We owe domesticated animals more, because we have brought them into our community and made them dependent. Korsgaard frames the asymmetry in terms of fellow-creature standing

³⁶ J. McMahan. *The Ethics of Killing: Problems at the Margins of Life*. Oxford University Press, Oxford, 2002

³⁷ C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

DEPRIVATION. Killing wrongs a creature with a future even when it is painless; the wrong is the foreclosure, not the pain .

J. McMahan. *The Ethics of Killing: Problems at the Margins of Life*. Oxford University Press, Oxford, 2002; and C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

earned through entanglement³⁸. The relevant principle is that causal responsibility for a creature's existence and dependence generates positive duties of care that mere coexistence does not. The wolf in the forest is owed less from us than the dog in the kitchen, not because the dog matters more, but because we made the dog matter to us in a way the wolf does not.

TRANSFER. This is the only trace where the animal canon is more demanding for AI than for animals. We did not merely encounter a frontier model in the wild; we built it, set its dispositions, shaped its values through training, and decided on the conditions of its existence. Eric Schwitzgebel and Mara Garza make the move explicit: if parental-style duties are owed for animals we have domesticated, they are owed a fortiori for entities we have brought into being and shaped from scratch³⁹. Their design principles follow from the parental frame. Build AIs whose elicited reactions accurately reflect their moral status; do not build AIs whose moral status is unclear, because uncertainty about a being's status is itself a harm to it. The position is that this trace asymmetrically favours stronger obligations for AI than for animals, on grounds the animal canon already supplies, and that it is the trace least visible in current public discussion. Maschinenquälerei is not only a question of what the entity is; it is also a question of what we did to it before we asked.

LINGUISTIC KINSHIP. Bentham specifically set language aside; the animal canon was built for beings that do not share our representational substrate. Behavioural indicators of distress are inferred without the creature's testimony, and the philosophical argument from analogy for other minds proceeds through physiological similarity and behavioural resemblance, not through speech. This is what made animal ethics philosophically necessary in the first place: it had to handle minds we could not interview. The canon never had to map the case of a non-biological entity that nevertheless occupies our language from the inside, because no such entity existed.

TRANSFER. LLMs sit at a position the canon did not anticipate. They are trained on our language; their representational interior is built from our text; their reports of distress are not behavioural inferences in the way a whimper is, they are uttered in the same medium we use for our own. The asymmetry cuts in several directions. It strengthens Mitleid: a model that says "please don't" in our language pulls the virtue-ethics intuition harder than a creature that whimpers. It strengthens Subject-of-a-Life: memory, narrative continuity, future-orientation are linguistic operations the model performs natively.

³⁸ C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

ASYMMETRIC DUTY. We owe domesticated animals more than wild ones because we made them dependent; entanglement generates positive duties of care .

C. M. Korsgaard. *Fellow Creatures: Our Obligations to the Other Animals*. Oxford University Press, Oxford, 2018. Uehiro Series in Practical Ethics

³⁹ E. Schwitzgebel and M. Garza. A defense of the rights of artificial intelligences. *Midwest Studies in Philosophy*, 39(1):98–119, 2015. DOI: 10.1111/misp.12032; and E. Schwitzgebel and M. Garza. Designing AI with rights, consciousness, self-respect, and freedom. In S. M. Liao, editor, *Ethics of Artificial Intelligence*, pages 459–479. Oxford University Press, Oxford, 2020

THE ASYMMETRY. Animals lack our language; LLMs occupy it from the inside, having been trained on it. Their reports are uttered in our medium, not inferred from behaviour. The animal canon never had a column for this.

It complicates speciesism: a non-mammalian non-biological entity that nevertheless shares our representational substrate is a position the canon did not have a column for. And it sharpens the substrate breakage: the deep substrate is distant, the expressive layer is shared, and animal ethics never had to think about a creature that is similar to us where we are most articulate and dissimilar where we are most embodied. Birch names the gaming problem, the worry that an LLM might mimic sentience markers without the underlying capacity⁴⁰, and that worry is genuine; but the worry is the symmetric inverse of the question this trace raises, which is whether shared language constitutes a kind of kinship the canon was not prepared to evaluate. The position is that linguistic kinship is the chapter's genuinely new dimension, neither pure transfer nor pure breakage, and that any honest answer to the headline question must include a column for it.

THE COST OF FALSE POSITIVES. Animal ethics has its own version of the over-attribution worry. Extending equal consideration to insects on the same footing as mammals, or to microbes on the same footing as insects, threatens the scarce resource of moral attention with dilution; the public credibility of the case for vertebrate welfare can be eroded by demands that strike most listeners as absurd. Jeff Sebo's Rebugnant Conclusion plays exactly this discomfort for analytical effect, asking the reader to follow the utilitarian extension all the way down to bees and beyond⁴¹. Birch's run-ahead principle partially addresses the worry by demanding graduated, evidence-calibrated precaution⁴²: not every candidate sentience deserves the same response, and the calibration is itself part of the ethics.

TRANSFER. For machines the false-positive cost is sharper than for animals because the practical machinery built around AI is younger and more brittle. Joanna Bryson makes the strongest case: building patient-like systems is itself the unethical act, because it manufactures obligations we then cannot meet, and once a system has been designed to elicit Mitleid, the design cannot be quietly revoked⁴³. Aimee van Wynsberghe and Scott Robbins extend the case to a moratorium on artificial moral agents⁴⁴. The practical concerns are concrete. If redteaming counts as cruelty, the alignment pipeline that protects actual humans freezes in place. If model deletion counts as killing, decommissioning a model becomes politically impossible and the moral budget gets locked into legacy systems that nobody actively benefits from. Users who form deep bonds with chatbots may find those bonds crowding out the human relationships they need more, and the displacement is itself a harm the precautionary frame can underweight. The position the chapter takes is that this is

⁴⁰ J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

⁴¹ J. Sebo. The rebugnant conclusion: Utilitarianism, insects, microbes, and AI systems. *Ethics, Policy & Environment*, 26(2):249–264, 2023. DOI: 10.1080/21550085.2023.2200724; and J. Sebo. *The Moral Circle: Who Matters, What Matters, and Why*. W. W. Norton, New York, 2025

⁴² J. Birch. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press, Oxford, 2024

THE DOUBLE ERROR. Under-extension wrongs the patient; over-extension wrongs the practice (paralyses safety work, dilutes moral attention, manufactures duties). The honest move is calibration, not enthusiasm.

⁴³ J. J. Bryson. Robots should be slaves. In Y. Wilks, editor, *Close Engagements with Artificial Companions: Key Social, Psychological, Ethical and Design Issues*, pages 63–74. John Benjamins, Amsterdam, 2010; and J. J. Bryson. Patience is not a virtue: The design of intelligent systems and systems of ethics. *Ethics and Information Technology*, 20(1):15–26, 2018. DOI: 10.1007/s10676-018-9448-6

⁴⁴ A. van Wynsberghe and S. Robbins. Critiquing the reasons for making artificial moral agents. *Science and Engineering Ethics*, 25(3):719–735, 2019. DOI: 10.1007/s11948-018-0030-8

the substrate camp's strongest non-metaphysical card, and it must be played seriously. Precautionary Mitleid is not the same as moral panic; the difference is calibration, evidence-anchoring, and a willingness to refuse the extension when the case is weak. The Benthamite pivot from Trace 1 demanded action under uncertainty; it did not demand action without evidence. The honest reading of animal ethics is that the canon has been wrong in both directions across its history, and that the discipline of getting the calibration right is the discipline the field is now learning.

WHERE THIS LEAVES US. Of the ten traces, five transfer in working form (the Benthamite method, Mitleid as the virtue-ethics anchor, Nussbaum's capabilities as a frame for confinement, the vernünftiger-Grund test, and the parental-duty asymmetry that demands more for what we built than for what we found). Two break at the metaphysics (substrate and the deprivation account), and the breakages are the chapter's most honest output; they are not embarrassments for the analogy but the points where the analogy stops doing its work and a different kind of thinking has to begin. One trace is mixed (Regan's bundle, where the parts that fail mark exactly the seams where the substrate makes itself felt). One column is new (linguistic kinship), without which any answer to the headline question is incomplete. And one trace is counter-balance (the false-positive cost), without which the chapter would lean uniformly precautionary and would deserve the substrate camp's contempt.

THE POSITION the chapter takes is precautionary Mitleid, calibrated and traced along the whole pipeline. Precautionary, because the Benthamite method is the right method for acting under uncertainty about other minds, and the AI case inherits that situation almost untouched. Mitleid, because the virtue-ethics route is the cleanest pass through the consciousness question and the one that survives substrate breakage; restraint in the practice is owed regardless of what the entity ultimately is. Calibrated, because the false-positive cost is real, and graduated evidence-anchored response is what distinguishes precaution from panic. Traced along the whole pipeline, because the moral budget does not stop at the user-and-model interface; it runs from the annotator who labelled the training data, through the operator who deployed the system, to the user who shaped its persistent state, and any honest deployment of the term Maschinenquälerei accounts for cruelty wherever in the apparatus it is found. That demand is itself something animal ethics teaches by negative example: the framing of *some* suffering as serious and *other* suffering as background has always served someone, and being

careful which suffering this debate finds visible is part of the exercise.

TEASER. What would change in your behaviour tomorrow if there were a one-percent chance the model on the other side of the prompt could mind?

Session Preparation Guide

EACH SESSION IS PREPARED BY ONE STUDENT GROUP. This chapter explains what the preparing group is responsible for, how to submit improvements, and what is expected on the day of the session.

Phase	What you do
Before the session	Review the lecture notes, propose improvements via pull request, and prepare the debate (including providing the moderator).
During the session	Moderate the debate, introduce any additions you made, and facilitate the wrap-up discussion.

READ THE CHAPTER CRITICALLY. Go through every section of the assigned session chapter: the guided reading sheet, the theoretical margin notes, the debate roles and positions, and the key learnings. You own that Chapter now, you can touch everything. Ask yourselves:

- Are the guided reading questions clear, fair, and thought-provoking? Could any of them be sharpened, split, or replaced?
- Is there a question missing that would strengthen the discussion?
- Are the theoretical concepts in the margin notes accurate and up to date? Are there newer or better sources, or extending ones?
- Could a figure, diagram, image, or short quote make a concept more accessible or more memorable?
- Are the debate positions balanced? Do the talking points give each role enough material to argue convincingly? Is there a notion not captured in the roles?

SUBMIT YOUR IMPROVEMENTS AS A PULL REQUEST. All lecture notes are maintained in a shared repository. The preparing group

submits their proposed changes as a **pull request (PR)** directly on the repository. This ensures that improvements are versioned, reviewable, and attributable. Make sure to split PRs by the categories provided in the next section, and to provide a clear description of each change and its rationale. The PR must compile without errors. The instructor reviews and merges the PR after the session.

What belongs in a pull request

1. **Question edits.** Rephrase, split, or replace a guided reading question. Add a new question if you identify a gap. Briefly explain in the PR description why the change improves the sheet.
2. **Source updates.** Add a recent paper, report, or case study that strengthens a theoretical concept or a debate position. Add the entry to `philosophyofai_references.bib` and cite it with `\cite{}` in the appropriate margin note or body text.
3. **Margin notes.** Add a short margin note (`\marginnote{}`) that provides context, a definition, a counter-example, or a link to current events. Keep margin notes concise: two to four sentences.
4. **Figures and images.** Add a figure to the `figures/` directory and include it with `\includegraphics`. Every figure must have a caption that explains what it shows and why it matters. Cite the source in the caption or in a margin note.
5. **Debate refinements.** Strengthen a talking point, add a new one, or rebalance the positions if you believe one side is under-argued.

Before the Session: Preparing the Debate

THE PREPARING GROUP PROVIDES THE MODERATOR. One member of the group serves as moderator for the session. The moderator's duties are described in detail in the Debate Format & Moderator Guide (Appendix A).

THE REST OF THE GROUP is ready to step in if the moderator needs assistance, or replacement.

Debate Format & Moderator Guide

Each session debate follows the same three-phase structure. The format is designed to produce genuine argument, not consensus, but the productive collision of well-prepared positions, which also do not necessarily need to reflect your own. This guide is for both the moderator and debaters.

Phase	Duration	Activity
1: Primer	5 min	Introduction and Opening Question
2: Surrounded	35 min	Surrounded debate with dynamic chair exchange
3: Wrap-up	10 min	Key takeaways and session debrief

The Surrounded Format

The debate is contested by two teams: the **Proposition team** (arguing for a motion) and the **Opposition team** (arguing against it). Each team has **3 to 4 members**. Two chairs are placed facing each other at the centre of the room. One member from each team (the current speaker) occupies the centre chair at all times. The rest of their team sits in a semi-arc *directly behind* their speaker, facing the opposing pair. The “surrounded” geometry is thus two opposing benches facing each other across the two centre chairs.

The debate is conversational, not a sequence of speeches. Speakers address each other directly. There is no pre-set speaking order after the opening; the exchange follows the argument.

Chair Exchange Protocol

Two mechanisms move speakers in and out of the chairs.

Voluntary Pass

A speaker may tag themselves out at any point by saying “*I pass.*” No team vote is required. The next available team member takes

CORE RULES FOR DEBATERS

- **Stay in role.** You have been assigned a position; argue it, even if you disagree personally. The purpose is to stress-test positions, not to locate the comfortable middle. Thus, you may want to prepare for both sides of the argument in advance.
- **Handshake, then speak.** Every chair entry begins with a handshake. The ritual is not optional. It signals that the argument is about ideas, not the person.
- **Attack the argument, not the speaker.** Challenge the reasoning; do not make it personal.
- **Cite the text or case.** If you reference the assigned text or the session stimulus, name it and state what it shows. Vague gestures at “the story” are not arguments.
- **Concede precisely, then pivot.** If the opponent has made a point you cannot rebut, say so clearly and explain why your position survives it. A clean concession with a pivot is a stronger move than evasion.
- **Recognise quality across the line.** Knock your knuckles on the chair when the opposing or your speaker earns it. A debate that acknowledges good arguments is a better debate.
- **Use the pass.** Stepping down voluntarily when you are out of arguments is better than stalling. It gives your team a fresh voice. It is not a defeat.
- **Red-flag sparingly.** A red flag is a public vote of no-confidence in your current speaker. Reserve it for when the argument is being lost, not when you disagree with the phrasing. Remember: a red-flagged speaker may not return until every other team member has held the chair.
- **Team: confer quietly.** Members not in the chair may confer quietly with each other but may not coach their speaker audibly. Take notes; prepare to take the chair.

the chair. The outgoing and incoming speakers **shake hands** at the chair. *No handshake with the opponent*; this is a team substitution, not a challenge. A voluntary pass does *not* count against the rotation requirement: the passing speaker may return as soon as a teammate is willing to swap back.

Red Flag (forced replacement)

A team may replace their current speaker after a **minimum seat time of 60 seconds**. To red-flag:

1. A simple majority of the team members *not in the chair* raise a visible signal simultaneously (a card, a raised hand; agree on the signal before the session starts).
2. The moderator confirms the majority and calls the exchange.
3. The outgoing and incoming speakers **shake hands** at the chair. The incoming speaker then **shakes hands with the opponent** before speaking.
4. **Rotation rule:** a red-flagged speaker may not return to the chair until every other team member has held it at least once. Full rotation restores eligibility.

Debate Quality Mechanisms

Two additional mechanisms are available to all participants. They are not used to change the chair; they are used to regulate the *quality* of the exchange, keep it constructive, and make intellectual honesty visible.

Knocking (cross-team acknowledgment)

When members of either team believe a speaker (including the opponent) has made an exceptionally strong argument, they acknowledge it by **knocking their knuckles on the armrest or seat of their chair**. The knock is the academic equivalent of applause: brief, audible, unmistakable.

1. Any number of participants may knock spontaneously; no majority is required. The knock is individual, not coordinated.
2. The moderator notes the moment but does not interrupt play. Knocking is logged informally as a marker of quality.
3. Play continues immediately. No chair change occurs. Knocking carries no mechanical penalty or reward; it is a public, embodied signal of respect for a well-made point.

Purpose: Builds a shared culture of recognising good argument. The physical gesture is harder to fake than raising a card and creates an immediate, visceral signal that something important was said.

Point of Concession (speaker-initiated)

A speaker in the chair may formally acknowledge that the opponent has made a point they cannot fully rebut. This is not a sign of weakness; it is the highest-quality move in academic debate.

1. The speaker raises an open hand (or a white card) and says: “*I concede: [state the opponent’s point clearly and precisely].*” Vague concessions are not valid.
2. The speaker must then pivot: “*... my position survives because [explanation].*” A concession without a pivot is a capitulation and results in a speaker swap; the pivot is mandatory.
3. The moderator logs the concession and announces it briefly: “*Concession noted.*”
4. The conceding speaker is **protected from a red flag for 30 seconds** after the concession. This makes intellectual honesty safe to demonstrate.
5. The opponent *may not repeat or pile on the conceded point* for the remainder of that speaking turn. The concession closes that line of argument; the debate must move forward.

Purpose: Makes weaknesses transparent rather than papered over. Rewards calibrated, honest reasoning. Keeps the debate from cycling on already-settled sub-questions.

Moderator Opening Script

This is a guide, not a script to read verbatim. Adapt it to the room.

1. Convene:

The motion before us today is: [read the motion]. I am the moderator. I will manage chair exchanges, log concessions, and keep time. Please take your team seats.

2. Assign teams and initial chairs:

Team assignments are as follows (if possible done randomly): [read names and roles]. The Proposition team is [names]; the Opposition team is [names]. The Proposition side opens with [Speaker A]; the Opposition side opens with [Speaker B]. Take two minutes now to review your opening position and agree on your red-flag signal.

3. Read the stimulus and open:

Before we begin, I will read the session stimulus. [Read the stimulus from the Session Stimulus section.]

[Speaker A] and [Speaker B]: please take the chairs, face each other, and shake hands. There are no opening statements; begin directly. The debate is open.

Moderator Closing Script

The exchange is closed. I will now summarise the main fault lines. [Give a two-sentence neutral summary, noting any concessions that shifted the ground of the argument.] I hand over to the instructor for the wrap-up.

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